

Chi-Yao (Trey) Huang

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SUMMARY

Ph.D. researcher in representation learning, 3D perception, and world models. My research focuses on interpretable and controllable latent-space learning for embodied AI, connecting spatial perception, semantic understanding, and reasoning through structured visual representations. Previously a Senior R&D Engineer at HTC VIVE, where I developed core VO/SLAM tracking systems deployed across HTC VIVE's major VR/AR product lines.

EDUCATION

- Arizona State University** Tempe, AZ
• *Ph.D. in Computer Science, Advisor: Prof. Yezhou Yang* 2023 – Present
M.S. in Robotics and Autonomous Systems 2021 – 2023
- National Taiwan University** Taipei, Taiwan
• *M.S. in Mechanical Engineering* 2015 – 2017
- National Sun Yat-Sen University** Kaohsiung, Taiwan
• *B.S. in Mechanical Engineering* 2010 – 2014

SELECTED PUBLICATIONS

- **Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning (ICLR 2026):** Proposed an orthogonal latent-space construction framework that decomposes task-specific factors into structured subspaces to reduce representation collapse in multi-task learning. Evaluated on ShapeNet and MPIIGaze, showing improved task separation, interpretability, and multi-task generalization.
([project](#) | [paper](#) | [code](#) | [talk](#))
- **VOCAL: Visual Odometry via ContrActive Learning (WACV 2026):** Proposed a contrastive representation learning framework for visual odometry that organizes latent representations according to camera motion. Evaluated on KITTI, demonstrating interpretable motion-aware latent structure for pose estimation.
([project](#) | [paper](#) | [talk](#) | [demo](#))

EXPERIENCE

- Arizona State University** Tempe, AZ
• *Ph.D. Researcher, Computer Vision and Robotics – Prof. Yezhou Yang* Aug 2021 – Present
 - Developing interpretable and controllable latent-space learning frameworks for representation learning, 3D perception, and embodied AI.
 - Researching structured visual representations that connect spatial perception, semantic understanding, and reasoning for world-model-oriented AI systems.
 - Designed contrastive learning and orthogonal latent-space methods to improve representation structure, task separation, and motion-aware visual understanding.
 - Evaluated representation learning methods on visual odometry, pose estimation, gaze estimation, and 3D perception tasks.
 - Supported by Toyota Research Institute of North America (TRINA).
- HTC VIVE, Advanced and Creative Team** New Taipei, Taiwan
• *Senior R&D Engineer, Team Lambda – VR/AR Tracking and Spatial Computing* Sep 2017 – Feb 2021
 - Developed core VIO/SLAM tracking systems deployed across HTC VIVE's major commercial VR/AR product lines, including VIVE COSMOS, VIVE FOCUS 3, VIVE XR Elite, and VIVE FLOW.
 - **VIVE COSMOS:** Architected a multi-camera tracking system for consumer VR, achieving sub-millimeter tracking precision in controlled evaluation; developed a high-performance “mothership” SLAM system. ([product](#) | [demo](#))
 - **VIVE FOCUS 3:** Led prototyping for a flagship MR system and optimized a tightly coupled visual-inertial SLAM pipeline on resource-constrained embedded platforms, achieving a 4× runtime speed-up. ([product](#))
 - **VIVE XR Elite:** Engineered low-latency VIO and obstacle avoidance systems for mixed-reality interaction; designed core data structures for real-time spatial computing. ([product](#))
 - **VIVE FLOW:** Designed an inside-out tracking algorithm for limited-FoV sensors and developed a prototype motion-tracking pipeline integrating real-time hand tracking. ([product](#))
 - **Spatial Mapping:** Built voxel-based obstacle mapping and semantic mapping modules for VR safety and real-world interaction.

- **NTU Robotics Lab, National Taiwan University** Taipei, Taiwan
Graduate Researcher and Vice System Manager – Prof. Han-Pang Huang *Sep 2015 – Jun 2017*
 - Conducted research in SLAM, 3D reconstruction, dense mapping, and robot motion planning.
 - Managed robotic equipment including robotic arms and mobile robots, and developed orientation materials for lab members.

SELECTED PROJECTS

- **Aerial Pose Estimation – Toyota Research Institute of North America:** Developed a UAV pose estimation framework using IR and IMU sensor fusion for high-altitude drone perception under challenging lighting and weather conditions. ([demo](#))
- **3D Landmark Reconstruction for Robot Localization and Mapping:** Implemented a learning-based 3D reconstruction pipeline that jointly optimizes camera trajectories and object-level semantic poses for robot localization. ([demo](#))
- **3D Reconstruction and Path Planning with Signed Distance Function:** Fused dense image information with sparse feature points for efficient SLAM; implemented voxel hashing for dense mapping and integrated the reconstruction module with robot path planning.
- **Humanoid Robot Control:** Developed bipedal walking patterns on non-planar surfaces by optimizing ZMP trajectories and implemented localized A* planning for obstacle navigation. ([demo](#))

HONORS, AWARDS & GRANTS

- **Research Grant, Toyota Research Institute of North America** 2023 – Present
- **Ira A. Fulton Schools of Engineering Fulton Fellows Award, Arizona State University**
- **Student with Distinction, Arizona State University**

SKILLS

- **Programming:** Python, C/C++, CUDA, JavaScript, MATLAB
- **Machine Learning:** Contrastive learning, representation learning, multi-task learning, generative models, foundation models, world models
- **Computer Vision and Robotics:** Visual odometry, SLAM, 3D reconstruction, pose estimation, bundle adjustment, sensor fusion, path planning, aerial robots
- **Tools and Frameworks:** PyTorch, OpenCV, Ceres, Eigen, ROS, Docker, Git, OpenGL, SIMD optimization, CEVA/Hexagon DSP

LANGUAGES

- English (Fluent), Mandarin (Native)